

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No Cummins® service tools required.

Additional Service Items

- Coolant hose(s)
- Container for coolant
- Thermometer
- Seal driver
- Heated water.

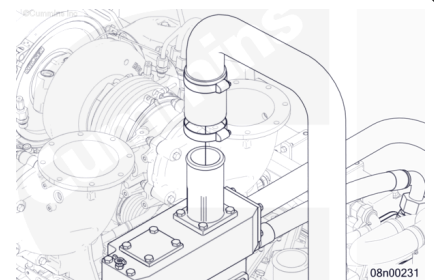
Leak Test

Note : This test should be performed on cold engine **only**. If coolant temperature reaches opening threshold of LTA thermostat, results will **not** be valid.

Engine thermostat and thermostat seal **must** operate properly in order for engine to operate in most efficient heat range. Overheating or overcooling will shorten engine life.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

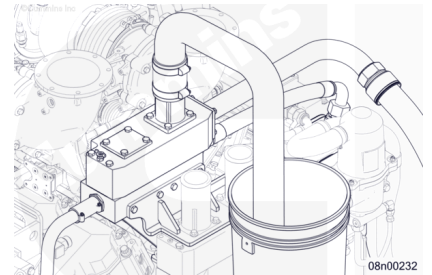


⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Remove the radiator hose from the LTA thermostat housing outlet.

Connect a hose to LTA outlet. The hoses **must** be long enough to reach a remote, dry container. Put the end of the hose in a dry container.

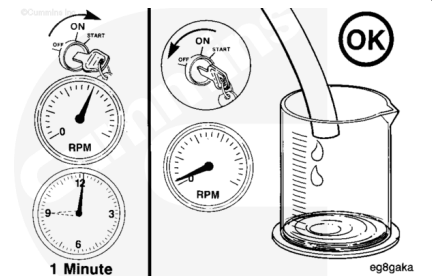


Operate the engine at rated rpm for one minute.

Shut the engine OFF.

Measure the amount of coolant collected in the container.

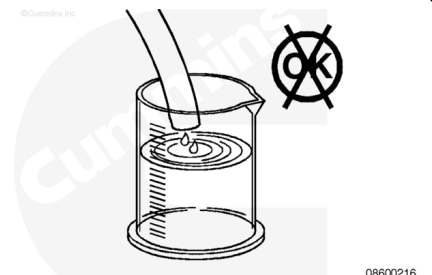
The amount of coolant collected **must not** be more than 100 cc [3.3 oz].



If more than 100 cc [3.3 oz] of coolant is collected, the thermostat or thermostat seal is leaking.

Remove the thermostats and replace the seals.

Test the thermostats for proper operation.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

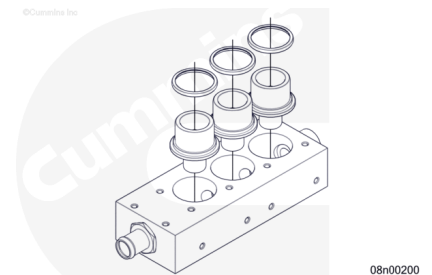
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/57/57-008-018.html>).
- Remove the LTA thermostat housing. Refer to Procedure 008-133 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-133.html>)

Remove

Remove the LTA thermostats.

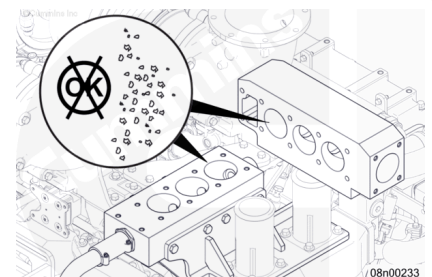
Remove the LTA thermostat seals.



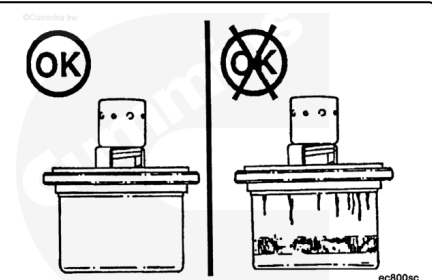
Clean and Inspect for Reuse

Inspect the LTA thermostat housing and support for debris and corrosion. Refer to Procedure 008-133 in Section 8 (</qs3/pubsys2/xml/en/procedures/57/57-008-133.html>).

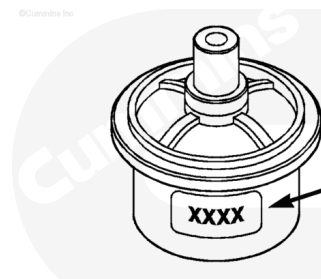
Clean the components as necessary.



Check the LTA thermostat for wear or damage. If the barrel of the thermostat is worn or fretted, it **must** be replaced.



Check to see at what temperature the LTA thermostat is designed to open. The rated opening temperature is marked on the side of the thermostat.



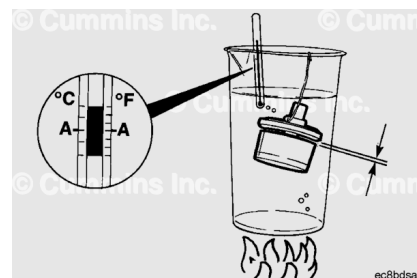
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Suspend the LTA thermostat and thermometer in a container of water. Do **not** let the thermostat or the thermometer touch the side of the container.

Heat the water to check the temperature when the LTA thermostat begins to open.

Compare the results with the values in Cooling System specifications. Refer to Procedure 018-018 in Section V (</qs3/pubsys2/xml/en/procedures/57/57-018-018.html>).

Replace the LTA thermostat if it does **not** meet specifications.

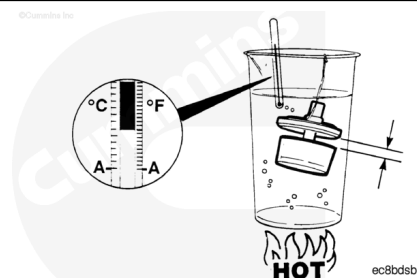


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Continue to apply heat to check the temperature when the LTA thermostat is open fully.

Compare the results with the values in Cooling System specifications. Refer to Procedure 018-018 in Section V (</qs3/pubsys2/xml/en/procedures/57/57-018-018.html>).

Replace the thermostat if it does **not** meet specifications.



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Remove the container from the heat.

Check to see if the LTA thermostat returns to the CLOSED position.

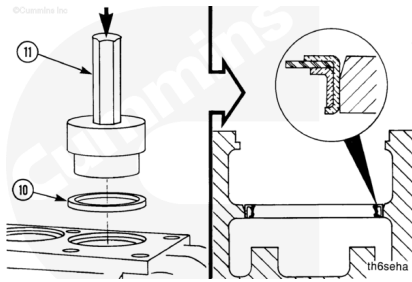
Replace the LTA thermostat if it does **not** close.



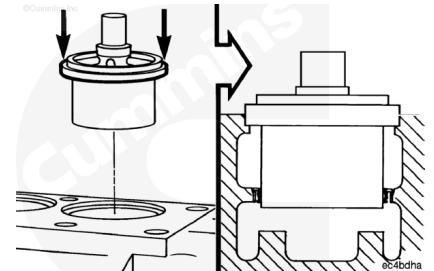
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Install

Install the LTA thermostat seals into the LTA thermostat housing. Use a mallet and suitable seal driver.



Push on the outer rim to install the LTA thermostat.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the LTA thermostat housing. Refer to Procedure 008-133 in Section 8 (</qs3/pubsys2/xml/en/procedures/57/57-008-133.html>).
- Fill the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/57/57-008-018.html>).
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to 85°C [185°F] coolant temperature. Check for leaks.